

CLOUD COMPUTING

Cloud computing is the delivery of computing services like **storage, servers, networking, databases, and software** over the **internet** on a **pay-as-you-use** basis.

Users do not need to manage physical hardware or software infrastructure.

Problems with Traditional IT Architecture

Traditional IT systems face several limitations:

- High cost for data centres and hardware
- Maintenance, cooling, and power expenses
- Limited scalability
- Location dependency
- Difficult disaster recovery

Why Cloud Computing?

Cloud computing solves traditional IT problems by offering:

- No upfront investment
- On-demand resource availability
- Easy scalability
- Reduced operational cost
- High availability and disaster recovery

Key Features of Cloud Computing

- **High Scalability** – Resources can be increased or decreased easily
- **Flexibility** – Access services anytime, anywhere
- **Cost Reduction** – No need to buy infrastructure
- **High Security** – Advanced security provided by vendors
- **High Availability** – 99.99% uptime
- **Disaster Recovery** – Fast recovery from failures

Cloud Deployment Models

A cloud deployment model defines **who owns the cloud, who can access it, and how secure it is.**

Types of Cloud Deployment Models:

1. Public Cloud
2. Private Cloud
3. Community Cloud
4. Hybrid Cloud

1. Public Cloud Model

Public Cloud resources are **owned and managed by third-party providers** and delivered via the internet.

Key Characteristics:

- Pay-per-use model
- Highly scalable
- No infrastructure management
- Suitable for large-scale applications

Advantages of Public Cloud:

- No infrastructure management
- Highly scalable
- Cost-effective
- High availability
- Ideal for startups and testing environments

Examples: Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), IBM Cloud, Oracle Cloud

2. Private Cloud Model

Private Cloud is used by a **single organization** and offers **high security**.

Limitations: Expensive, Requires maintenance, Limited scalability

3. Community Cloud Model

Shared by organizations with **similar requirements** (e.g., banks, universities).

Benefits:

- Cost sharing
- Better security than public cloud

4. Hybrid Cloud Model

Combination of **Public + Private Cloud**.

Advantages:

- High flexibility
- Better security
- Scalable and reliable